

Problem Set 1

Data Visualisation for Social Scientists

Due: January 28, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Wednesday January 28, 2026. No late assignments will be accepted.

Roll Call Votes in the European Parliament

Data Manipulation

First, you need to download data from the first six elected European Parliaments on each MEP and how they voted in each recorded roll-call vote.

1. Load these datasets into your global environment:

- `mep_info_26Jul11.xls` (MEP characteristics, EP1–EP5)
- `rcv_ep1.txt` (EP1 roll-call votes)

```
1 MEP_data <- read_csv("mep_info_EP1.csv")  
2 EP1_data <- read_csv("rcv_ep1.txt")
```

2. Briefly describe (2–3 sentences each) the unit of analysis and key variables in each of these two datasets.

The unit analysis for the EP 1 dataset is the way of which each MEP voted. The key variables are MEP ID, Vote, Vote ID, and Country. The unit of analysis for the MEP dataset is each MEP in the term. The key variables are MEP ID, EPG, National Party, NOM D1, NOM D2, and country. This dataset is more of the attributes of the MEP while the EP 1 is more focused on the vote preference of each MEP.

3. The `rcv.ep1` data are in a wide format, with $V_1, V_2, \dots, V_n$ as separate vote columns.

- Identify which columns are ID/metadata (*MEPID*, *MEPNAME*, *MS*, *NP*, *EPG*) and which columns are vote decisions ($V_1 \dots V_n$). Tidy the voting data such that each row/observation is a single vote for a single MEP.
- Create a summary table of counts of decision categories (e.g. Yes/No/Abstain/Present but did not vote/Absent) across all votes.

```

1 #Pivoting data frame from wide to long
2 EP1_data <- EP1_data %>%
3   pivot_longer(
4     cols = starts_with("V"),
5     names_to = "vote_id",
6     values_to = "vote"
7   )
8
9 #Assigning the numbers to their vote type for future calculations
10 EP1_data <- EP1_data %>%
11   mutate(
12     vote = recode(vote,
13                   '0' = "Absent",
14                   '1' = "Yes",
15                   '2' = "No",
16                   '3' = "Abstain",
17                   '4' = "Present but did not vote",
18                   '5' = "Not an MEP"
19     )
20   )
21
22 #Creating a summary table with the counts for each vote type
23 EP1_summary <- EP1_data %>%
24   count(vote, name = "n")

```

For this section of manipulation, I converted the EP 1 data into a long format so I could have access to every vote outside of the column format. The I used the numeric values for the vote and assigned them to the correlating vote decision based on the what the source said. Finally, I made a table with the counts for each voting decision.

4. Construct a new dataset that combines MEP-level information with their vote decisions from EP1 in long format (from part 3). Check for missingness.

```

1 #wanted to change this to numeric before combining data sets
2 MEP_data <- MEP_data %>%
3   mutate(
4     NOM_D1 = as.numeric(as.character('NOM-D1')),
5     NOM_D2 = as.numeric(as.character('NOM-D2'))
6   ) %>%
7   filter(
8     !is.na(NOM_D1),
9     !is.na(NOM_D2)
10  )
11
12 #4
13
14 #I want to connect the tables through the MEP ID values and EP Group
15 #but the column names are different so the code below is
16 #renaming the column to match
17 colnames(MEP_data)
18 names(MEP_data)[names(MEP_data) == "MEP id"] <- "MEPID"
19 names(MEP_data)[names(MEP_data) == "EP Group"] <- "EPG"
20
21 #Combining the two data sets
22 combined_data <- merge(MEP_data, EP1_data)
23 sapply(combined_data, function(x) sum(is.na(x))) #checking that theres no
    NAs

```

For this section of code, I had run it without converting the NOM D1 and NOM D2 columns into numeric values and found that later on it made some of the calculations NA or impossible. So, before merging the two datasets I converted these columns. Another issue I found while merging the datasets was that columns with the same observations were named differently between the two datasets. Therefore, I changed the names on the MEP dataset to match the EP1 dataset. Finally, I merged the two datasets and checked for NAs. There were some in the ...8 column, but I found that column to be unimportant to the tasks at hand.

5. Compute, for each EP group in EP1:

- The mean rate of Yes votes (Yes over Yes+No+Abstain) across all roll calls.
- The mean abstention rate.
- The mean vote preferences along the two contested dimensions (NOM-D1 and NOM-D2).

```

1 #Mean vote rates for each EP group
2 means_EPG <- combined_data %>%
3   group_by(EPG) %>%
4   summarise(
5     n_yes = sum(vote == "Yes", na.rm = TRUE),
6     n_no = sum(vote == "No", na.rm = TRUE),
7     n_abstain = sum(vote == "Abstain", na.rm = TRUE),

```

```

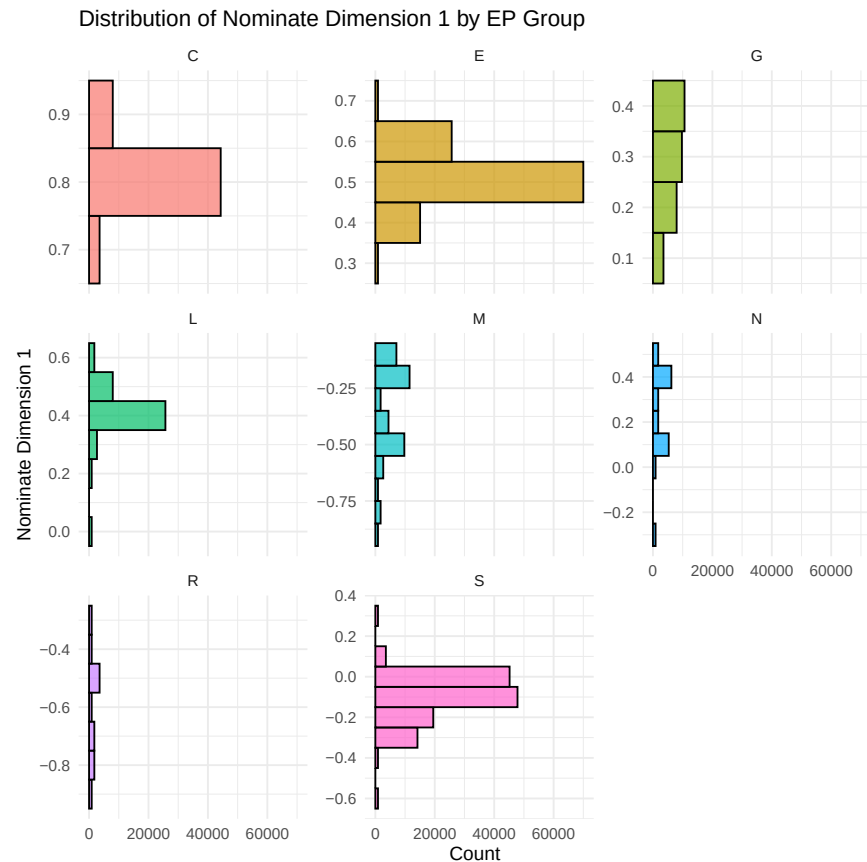
8     mean_yes_rate = n_yes / (n_yes + n_no + n_abstain),
9     mean_abstain_rate = n_abstain / (n_yes + n_no + n_abstain),
10    mean_nomd1 = mean(NOM.D1, na.rm = TRUE),
11    mean_nom2 = mean(NOM.D2, na.rm = TRUE),
12    .groups = "drop"
13 )

```

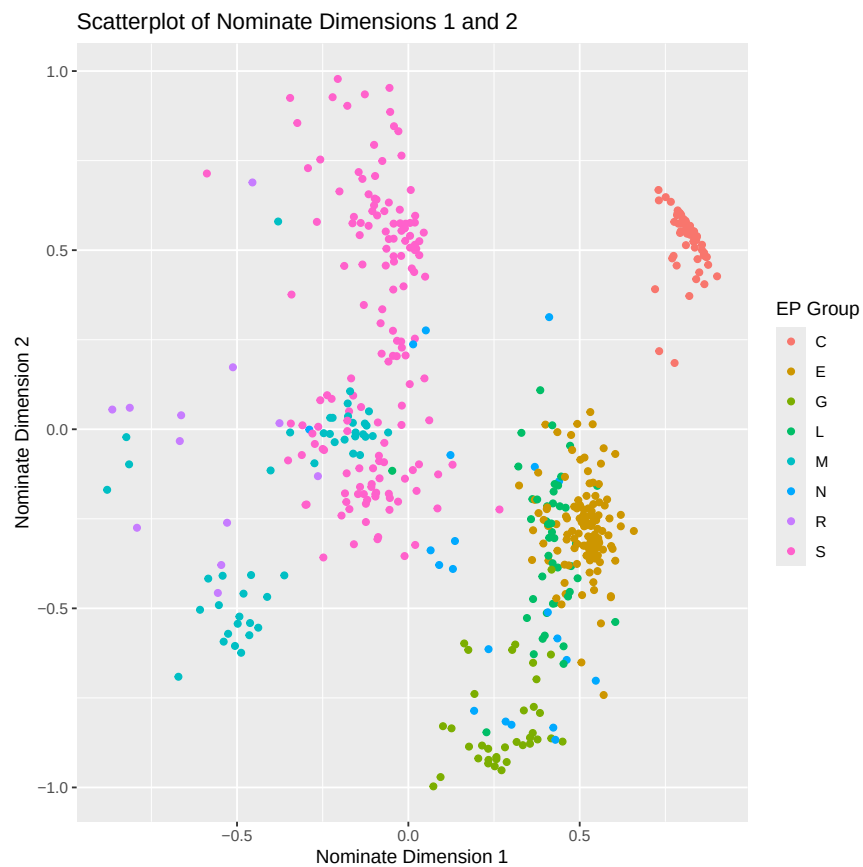
This section I created one table with all the mean values asked for in the information above. I found that when just calculating the means of Yes, No, and Abstain, the numbers didn't make sense since there was no count of them in this table based on EP group. Therefore, I calculated the vote counts by group in order to compute the mean. For the NOM D1 and NOM D2 data, I was able to section them based on group more easily.

Data Visualization

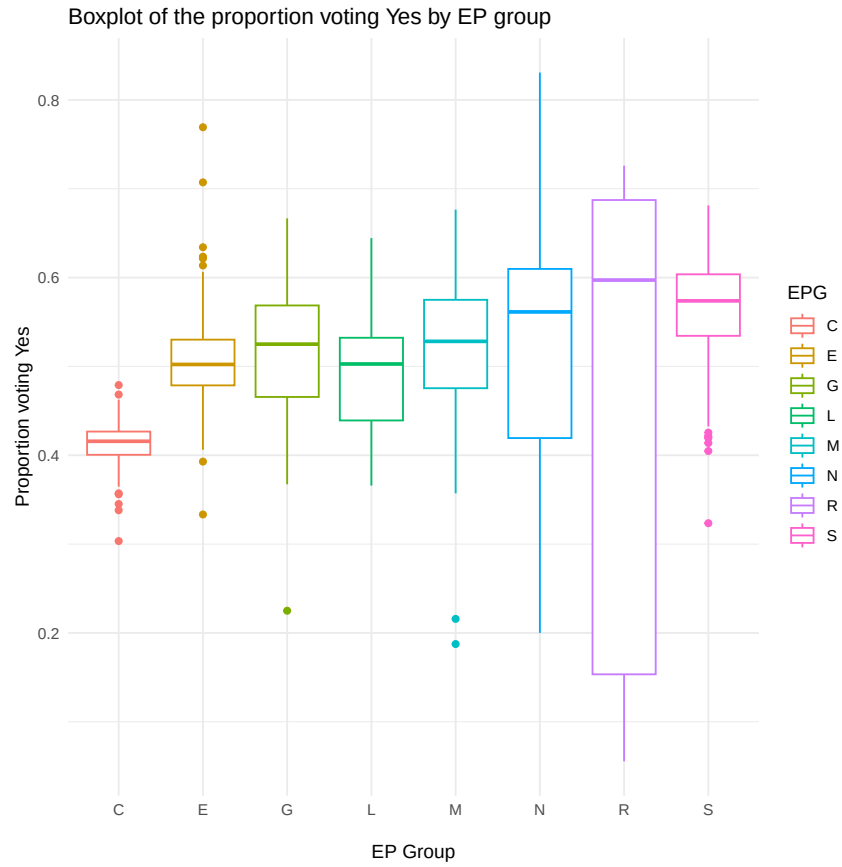
1. Plot the distribution of the first NOMINATE dimension by EP group, and explain any trends you see.



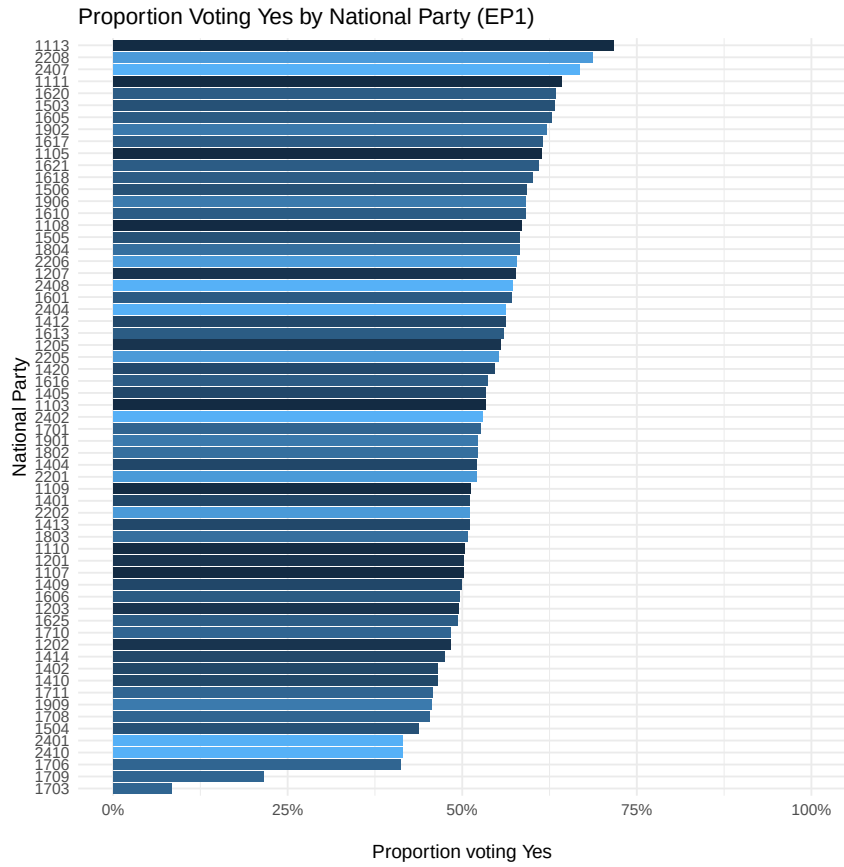
2. Make a scatterplot of *nomdim1* (x-axis) and *nomdim2* (y-axis), with one point per MEP and color by EP group.



3. Produce a boxplot of the proportion voting *Yes* by EP group to visualize cohesion.



4. Display the proportion voting *Yes* per year by national party using a bar plot.



5. For each EP group, calculate the average *Yes* share per year and plot a line graph.

N/A